
Your solutions need to be handwritten. After writing down your solutions, scan and compile those into a single pdf file. Then submit in the following link:

<https://forms.gle/aGYFFoSUWYCKzFUT6>

[The form will be accessible only from your G-Suite email account.]

Problem 1

Write down regular expressions for the following languages.

- a) $\{w \in \{0, 1\}^* : w \text{ does not end in } 101\}$ (2 points)
- b) $\{w \in \{a, b, c\}^* : w \text{ starts with } abba \text{ and ends in } bac\}$ (2 points)
- c) $\{w \in \{0, 1\}^* : w \text{ has length at least 3 and its third symbol is a } 0\}$ (2 points)
- d) $\{w \in \{0, 1\}^* : w \text{ contains an even number of } 0\text{'s or exactly two } 1\text{'s}\}$ (2 points)
- e) $\{w \in \{a, b\}^* : \text{the number of } a\text{'s in } w \text{ is 1 more than a multiple of } 3\}$ (2 points)

Problem 2

Construct deterministic finite automata for the following languages.

- a) $\{w \in \{0, 1\}^* : w \text{ contains exactly two } 0\text{'s}\}$ (2 points)
- b) $\{w \in \{0, 1\}^* : w \text{ contains an even number of } 0\text{'s and an odd number of } 1\text{'s}\}$ (2 points)
- c) $\{w \in \{0, 1\}^* : \text{the length of } w \text{ is even and } w \text{ contains } 0\text{'s at all the odd positions}\}$ (3 points)
- d) $\{w \in \{0, 1\}^* : w \text{ contains } 01^m0 \text{ as a substring where } m \text{ is divisible by } 3\}$ (3 points)
- e) $\{w \in \{0, 1\}^* : w \text{ contains } 0\text{'s and } 1\text{'s in alternate}\}$ (3 points)
- f) $\{w \in \{0, 1\}^* : \text{the last two symbols of } w \text{ are the same}\}$ (3 points)
- g) $\{w \in \{0, 1\}^* : w = 0^m1^n \text{ where } m \text{ is even and } n \geq 0\}$ (3 points)
- h) $\{w \in \{0, 1\}^* : w \text{ doesn't contain } 00 \text{ and } 11\}$ (3 points)
- i) $\{w \in \{0, 1, 2\}^* : w \text{ when interpreted in three base number system is divisible by nine}\}$ [Try using 3 states maximum] (3 points)
- j) [Bonus Question] Let $\Sigma = \{0, 1\}$. For $w \in \Sigma^*$, we denote by $h(w)$ the number of 1s in w . For example, $h(100101) = 3$.

Consider the following language over Σ . As always, $|w|$ is the length of the string w .

$$L = \{w : |w|^2 + h(w)^2 \text{ is two more than a multiple of three}\}$$

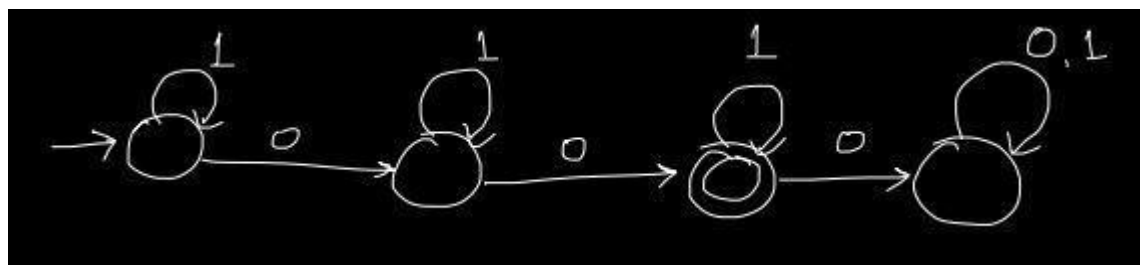
Give the state diagram for a DFA that recognizes L.

Problem 1

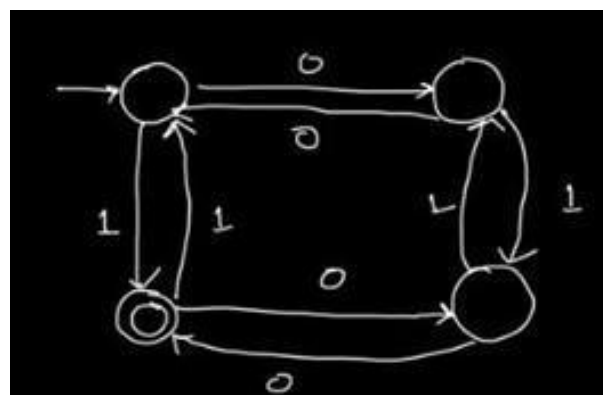
- a) $(0 \mid 1)^* (0 \mid 11 \mid 001) \mid \epsilon \mid 1 \mid 01$
- b) $abba (a \mid b \mid c)^* bac \mid abba$
- c) $(0 \mid 1) (0 \mid 1) 0 (0 \mid 1)^*$
- d) $(1^* 0 1^* 0 1^*)^* \mid 0^* 1 0^* 1 0^*$
- e) $(b^* a b^* a b^* a b^*)^* b^* a b^*$

Problem 2

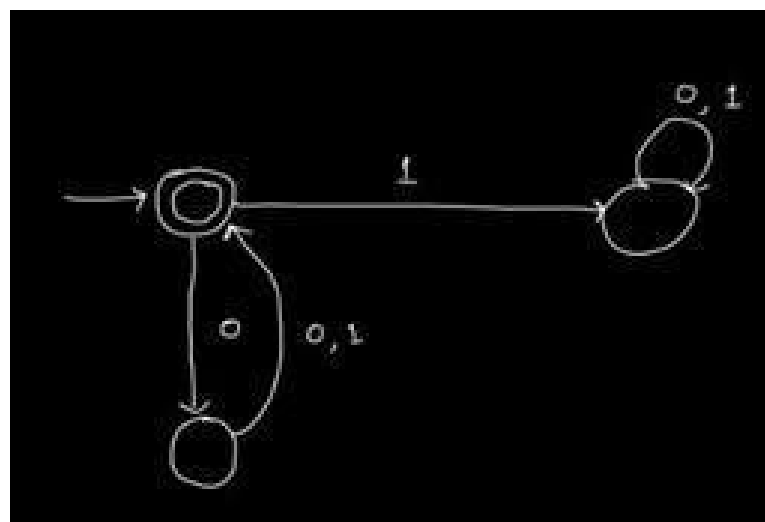
a)



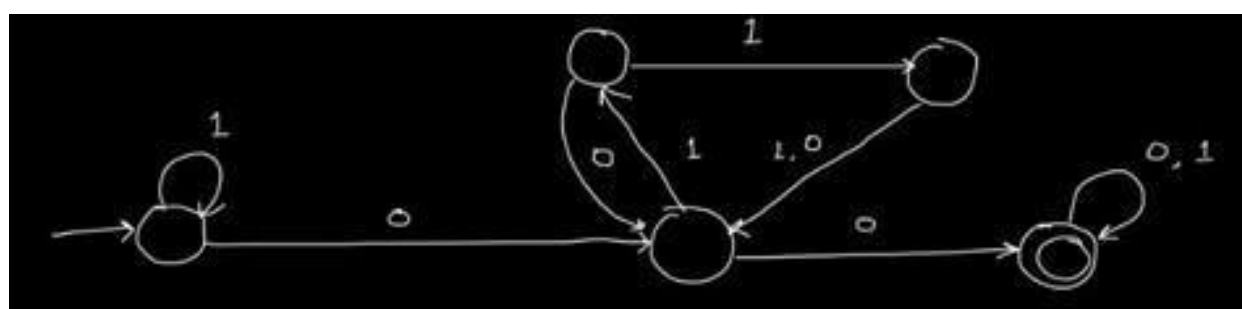
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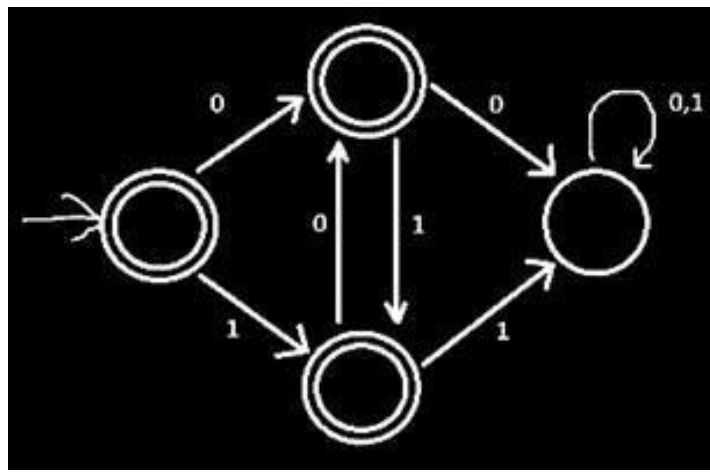
c)



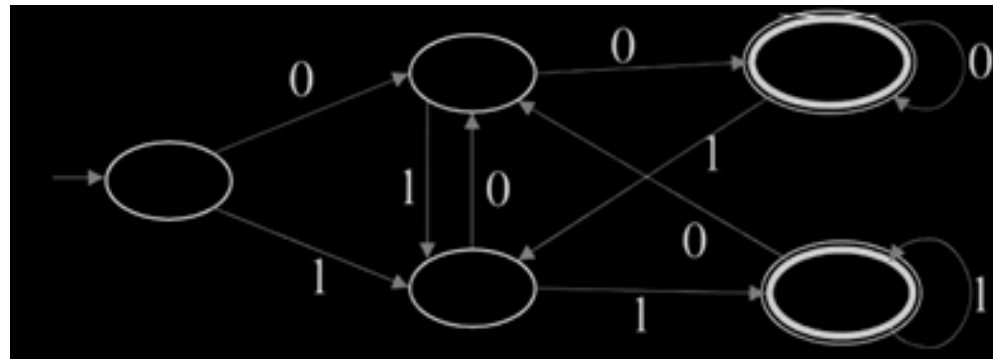
d)



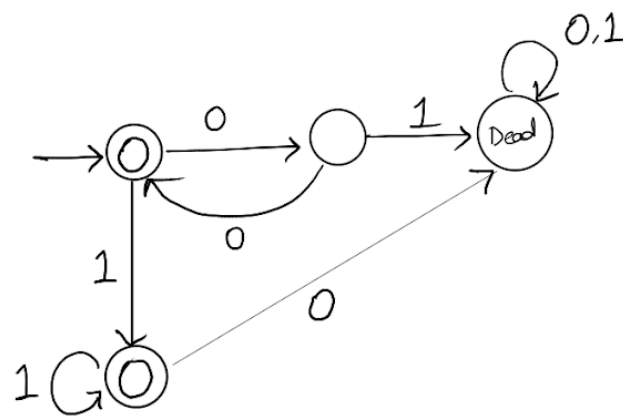
e)



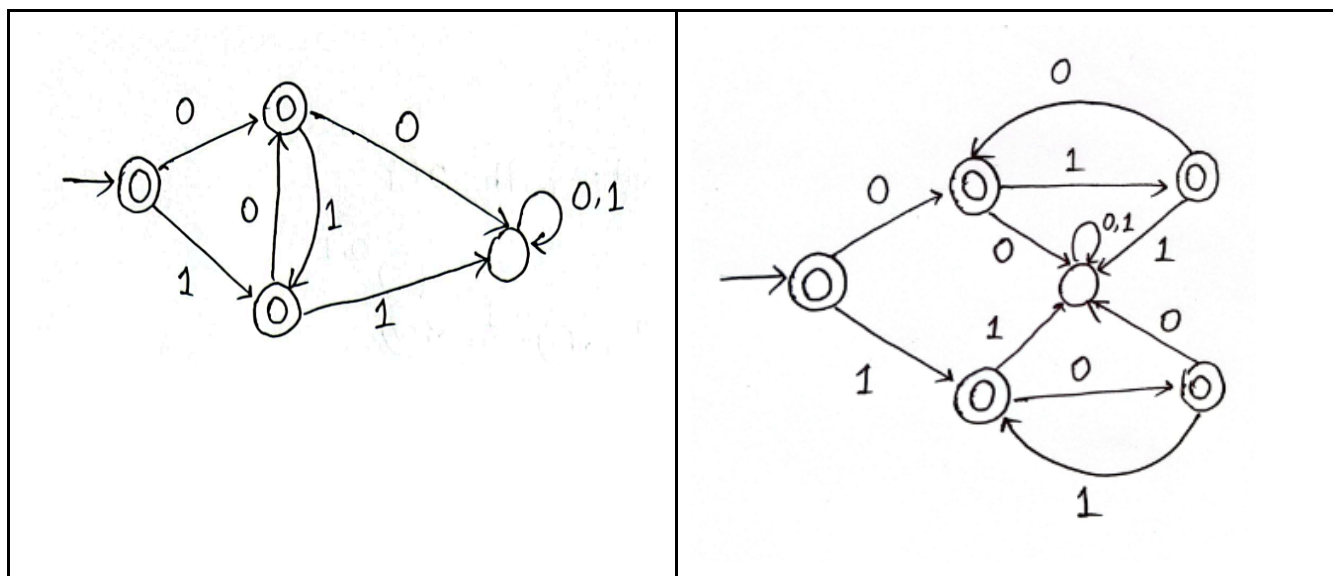
f)



g)

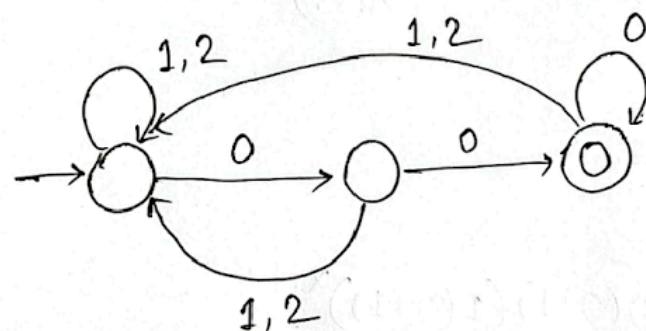


h)



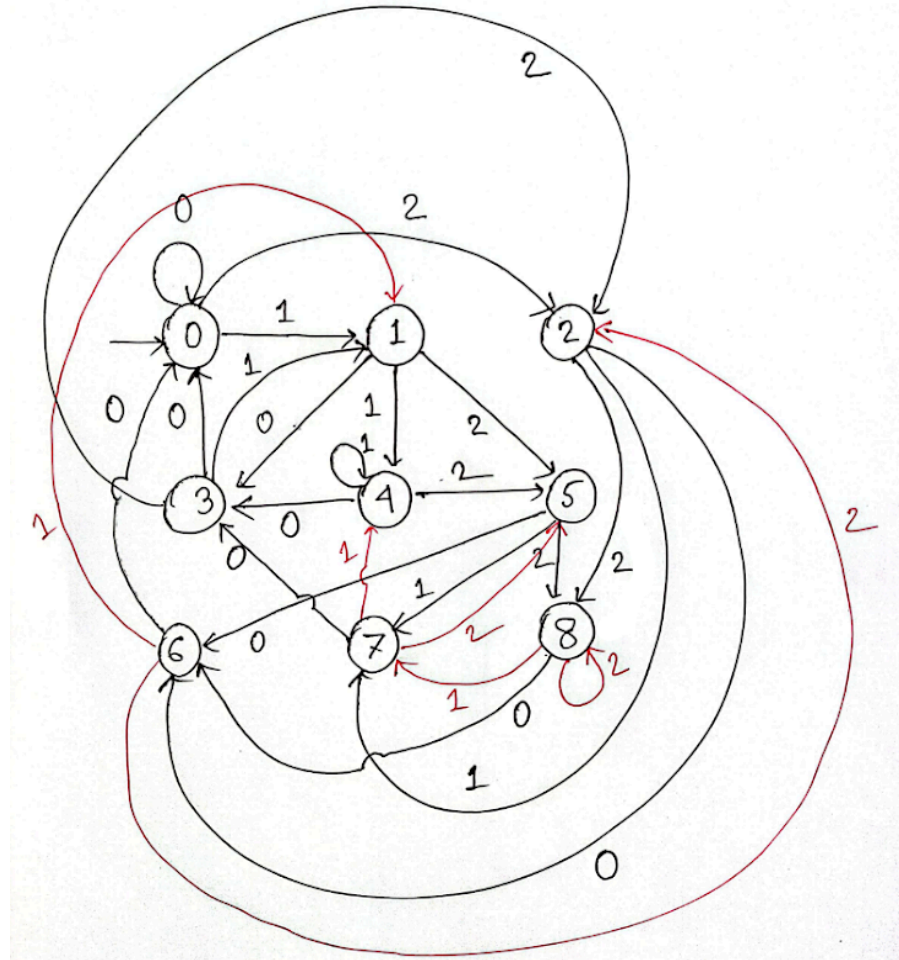
i)

In three base number system, the numbers which ends with 00, only those numbers are divisible by 9.

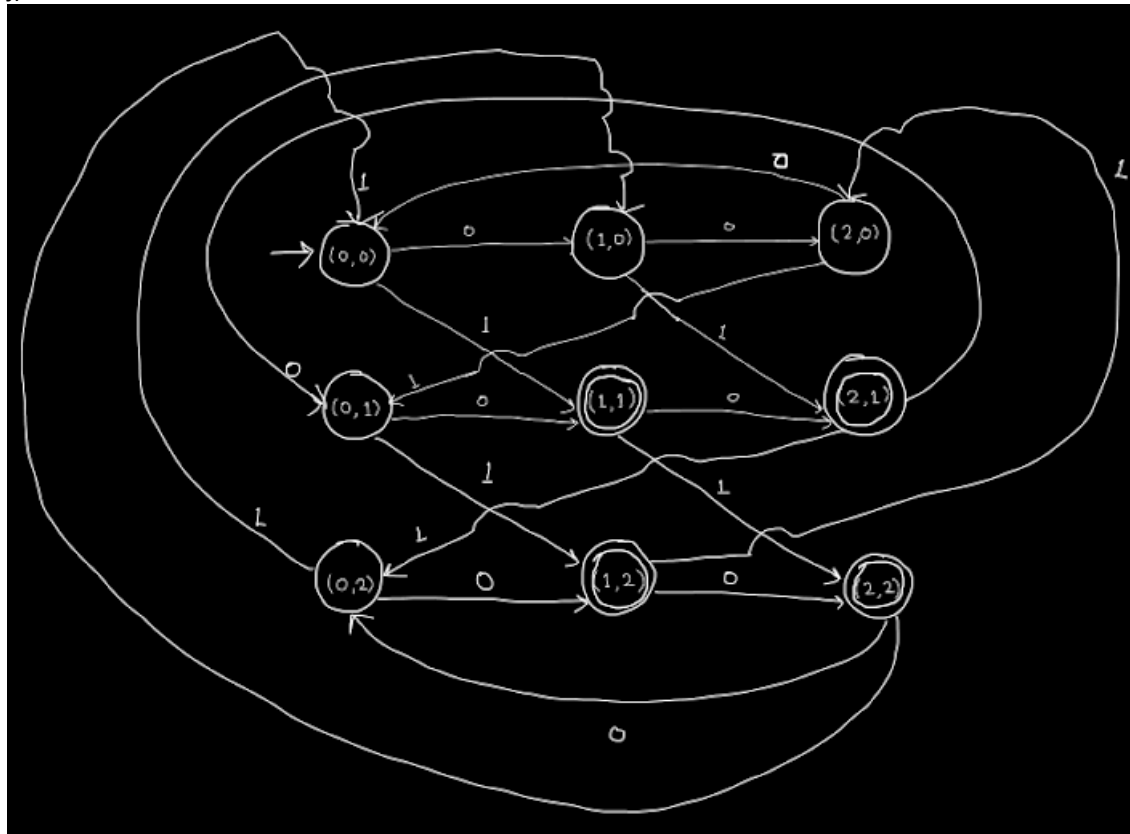


There is also a DFA with 9 states and 27 transitions - which is also correct.

The logic is - if there is a number x , in ternary representation, adding a 0 at the end will make the number $3x$, adding 1 at the end will make it $3x+1$ and for 2 it will become $3x+2$.



j)



A correct solution is anything equivalent to the above. The DFA diagram is a bit clunky due to there being a lot of states and transitions. However, the idea is simple: the states are indexed by a pair (i, j) where $0 \leq i, j \leq 2$, and i, j are the length and the number of 1s seen so far modulo 3. Upon seeing a new letter i goes up by 1, while j goes up by 1 only if the new letter is a 1.

The main observation is that $i^2 + j^2 = 2 \pmod 3$ iff $i = 1, 2$ and $j = 1, 2 \pmod 3$. This gives us four accepting states.

Any DFA for this language necessarily requires 9 states. So, any DFA with fewer states is wrong.

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Question 1 [20 Points]

Use the Pumping Lemma to demonstrate that the following languages are not regular.

- a) $L1 = \{w \in \{0, 1\}^* : 0^i 1^j \text{ where } i \leq j\}$
- b) $L2 = \{w \in \{a, b, c\}^* : a^i b^j c^{k+2} \text{ where } i = k \text{ and } i, j, k \geq 0\}$
- c) $L3 = \{w1\#w2 \text{ such that } |w1| = 2 \cdot |w2|, \text{ where } w1, w2 \in \{0, 1\}^*\}$
- d) $L4 = \{w \in \{a\}^* : a^{2^n} \text{ where } n \geq 0\}$
- e) $L5 = \{w10^i : w \in \{0, 1\}^*, i \geq 0 \text{ and } |w| \geq i\}$

Question 2 [20 Points]

Give a context-free grammar for each of the following languages.

Consider, $\Sigma = \{0, 1\}$.

- A. The language of strings that start with 1
- B. The language of strings of the form ww^R [w^R is the reverse of w]
- C. The language of strings that contain the substring 001
- D. The language $\{0^n 10^n \text{ where } n \geq 0\}$
- E. The language $\{0^i 1^j \text{ where } i \leq j\}$
- F. The language $\{1^i 01^j 01^{i+j} \text{ where } i, j \geq 0\}$
- G. The language $\{0^{3n} w 1^{2n} \text{ where } n \geq 0 \text{ and } w \text{ is a string that contains at least two } 0\text{'s}\}$
- H. The language $\{0^i 10^j \mid i \text{ is a multiple of three where } i, j \geq 0\}$
- I. The language $\{0^i 10^j \mid j = 2 + 3i \text{ where } i, j \geq 0\}$
- J. The language $\{1^i 01^j \mid j \text{ is a multiple of four or } i = 3 + 2j \text{ where } i, j \geq 0\}$

Question 3 [10 Points]

1. Suppose, you are given the following grammar. $\Sigma = \{0, 1\}$

$$S \rightarrow SS \mid 0S1 \mid 1S0 \mid 01 \mid 10$$

- a) Show that the grammar is ambiguous by demonstrating two different parse trees.
[Hint: String 011010]
- b) Find two strings of length six in the grammar above where exactly one parse tree is possible.

2. Suppose, you are given the following grammar. $\Sigma = \{a, b\}$

$$\begin{aligned} S &\rightarrow PE \\ P &\rightarrow aPa \mid bPb \mid a \mid b \\ E &\rightarrow aaE \mid abE \mid baE \mid bbE \mid \epsilon \end{aligned}$$

- a) Give a leftmost derivation for the string abababb and sketch the parse tree corresponding to that derivation.
- b) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (a)) for the same string abababb.

CSE331 Assignment 2 Solutions

Fall 2025

Question 1

a) $L1 = \{w \in \{0, 1\}^*: 0^i 1^j \text{ where } i \leq j\}$

Answer: Assume, for the sake of contradiction, $L1$ is regular. Then let p be the pumping length for $L1$. Now if we take the string: $w = 0^p 1^p \in L1$, then the length of w is $|w| = 2p \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L1$ for each $i \geq 0$.

Assume, $x = 0^\alpha$, $y = 0^\beta$ and $z = 0^{p-\alpha-\beta} 1^p$

Since $|xy| \leq p$, and the first p characters of w are all 0's, we can conclude that y consists of only 0's.

Then, for $i = 2$, $xy^2z = 0^\alpha \cdot 0^{2\beta} \cdot 0^{p-\alpha-\beta} 1^p = 0^{p+\beta} 1^p \notin L1$ since β can not be zero ($|y| > 0$).

We have excess 0's in xy^2z . Thus we get a contradiction! Hence, $L1$ is not a regular language.

b) $L2 = \{w \in \{a, b, c\}^*: a^i b^j c^{k+2} \text{ where } i = k \text{ and } i, j, k \geq 0\}$

Answer: Assume, for the sake of contradiction, $L2$ is regular. Then let p be the pumping length for $L2$. Now if we take the string: $w = a^p b c^{p+2} \in L2$, then the length of w is $|w| = 2p+3 \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L2$ for each $i \geq 0$.

Assume, $x = a^\alpha$, $y = a^\beta$ and $z = a^{p-\alpha-\beta} b c^{p+2}$

Since $|xy| \leq p$, and the first p characters of w are all a's, we can conclude that y consists of only a's.

Then, for $i = 2$, $xy^2z = a^\alpha \cdot a^{2\beta} \cdot a^{p-\alpha-\beta} b c^{p+2} = a^{p+\beta} b c^{p+2} \notin L2$ since β can not be zero ($|y| > 0$).

We have excess a's in xy^2z . Thus we get a contradiction! Hence, $L2$ is not a regular language.

c) $L3 = \{w1\#w2 \text{ such that } |w1| = 2 \cdot |w2|, \text{ where } w1, w2 \in \{0, 1\}^*\}$

Answer: Assume, for the sake of contradiction, $L3$ is regular. Then let p be the pumping length for $L3$. Now if we take the string: $w = 0^{2p} \# 1^p \in L3$, then the length of w is $|w| = 3p+1 \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L3$ for each $i \geq 0$.

Assume, $x = 0^\alpha$, $y = 0^\beta$ and $z = 0^{2p-\alpha-\beta} \# 1^p$

Since $|xy| \leq p$, and the first p characters of w are all 0's, we can conclude that y consists of only 0's.

Then, for $i = 2$, $xy^2z = 0^\alpha \cdot 0^{2\beta} \cdot 0^{2p-\alpha-\beta} \# 1^p = 0^{2p+\beta} \# 1^p \notin L3$ since β can not be zero ($|y| > 0$).

We have excess 0's in xy^2z . Thus we get a contradiction! Hence, $L3$ is not a regular language.

d) $L_4 = \{w \in \{a\}^* : a^{2^n} \text{ where } n \geq 0\}$

Answer: Assume, for the sake of contradiction, L_4 is regular. Then let p be the pumping length for L_4 . Now if we take the string: $w = a^{2^p} \in L_4$, then the length of w is $|w| = 2^p \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L_4$ for each $i \geq 0$.

Assume, $x = a^\alpha$, $y = a^\beta$ and $z = a^{2^p - \alpha - \beta}$

Then, for $i = 2$, $xy^2z = a^\alpha \cdot a^{2\beta} \cdot a^{2^p - \alpha - \beta} = a^{2^p + \beta}$.

If $xy^2z \in L_4$, then $2^p + \beta$ should be a power of 2 (obviously $2^p + \beta > p$). As β can not be zero, the immediate next power of 2^p is 2^{p+1} .

So, $2^p + \beta > 2^{p+1}$

$\Rightarrow 2^p + \beta > 2 \cdot 2^p$

$\Rightarrow \beta > 2^p$

But from condition 1,

$|xy| \leq p$

$\Rightarrow |y| \leq p$

$\Rightarrow \beta \leq p$

Thus we get a contradiction! Hence, L_4 is not a regular language.

e) $L_5 = \{w10^i : w \in \{0, 1\}^*, i \geq 0 \text{ and } |w| \geq i\}$

Answer: Assume, for the sake of contradiction, L_5 is regular. Then let p be the pumping length for L . Now if we take the string: $w = 0^p 10^p \in L$, then the length of w is $|w| = 2p + 1 \geq p$. So w can be split into xyz such that

1. $|xy| \leq p$,
2. $|y| > 0$ and
3. $xy^iz \in L$ for each $i \geq 0$.

Assume, $x = 0^\alpha$, $y = 0^\beta$ and $z = 0^{p-\alpha-\beta}10^p$

Since $|xy| \leq p$, and the first p characters of w are all 0's, we can conclude that y consists of only 0's.

Then, for $i = 0$, $xy^0z = 0^\alpha \cdot 0^{p-\alpha-\beta}10^p = 0^{p-\beta}10^p$

Now, $p - \beta \geq p \Rightarrow \beta \leq 0$

Therefore, $0^{p-\beta}10^p \notin L$ since $\beta > 0$. Hence, L_5 is not regular.

Question 2

A. $S \rightarrow 1P$

$P \rightarrow 0P \mid 1P \mid \epsilon$

Or, $S \rightarrow S0 \mid S1 \mid 1$

B. $S \rightarrow 0S0 \mid 1S1 \mid \epsilon$

C. $S \rightarrow P001P$

$P \rightarrow 0P \mid 1P \mid \epsilon$

D. $S \rightarrow 0S0 \mid 1$

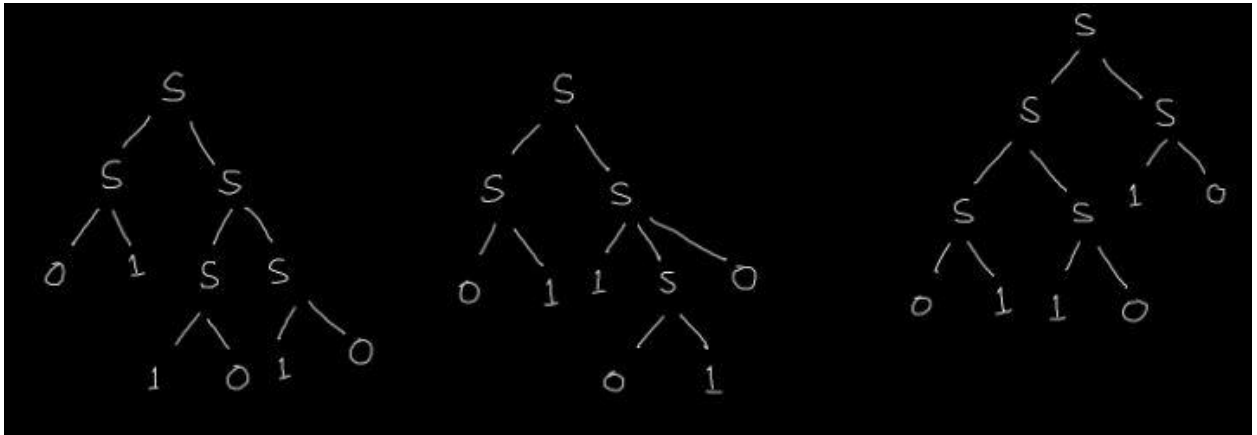
E. $S \rightarrow PS1 \mid \epsilon$

- $P \rightarrow \emptyset \mid \varepsilon$
Or, $S \rightarrow \emptyset S1 \mid Z$
 $Z \rightarrow Z1 \mid \varepsilon$
- F. $S \rightarrow 1S1 \mid \emptyset P$
 $P \rightarrow 1P1 \mid \emptyset$
- G. $S \rightarrow \emptyset \emptyset S11 \mid X \emptyset X \emptyset X$
 $X \rightarrow \emptyset X \mid 1X \mid \varepsilon$
Or, $S \rightarrow \emptyset \emptyset S11 \mid A \emptyset A \emptyset X$
 $A \rightarrow 1A \mid \varepsilon$
 $X \rightarrow \emptyset X \mid 1X \mid \varepsilon$
- H. $S \rightarrow R1P$
 $R \rightarrow \emptyset \emptyset \emptyset R \mid \varepsilon$
 $P \rightarrow \emptyset P \mid \varepsilon$
- I. $\emptyset^i 10^j = \emptyset^i 10^{2+3i} = \emptyset^i 10^2 \emptyset^{3i} = \emptyset^i \underline{1000} \emptyset^{3i}$
 $S \rightarrow \emptyset S \emptyset \emptyset \emptyset \mid 1 \emptyset \emptyset$
Or, $\emptyset^i 10^j = \emptyset^i 10^{2+3i} = \emptyset^i 10^{3i} \emptyset^2 = \emptyset^i \underline{1} \emptyset^{3i} \underline{\emptyset \emptyset}$
 $S \rightarrow P \emptyset \emptyset$
 $P \rightarrow \emptyset P \emptyset \emptyset \emptyset \mid 1$
- J. $S \rightarrow P \emptyset R \mid Q$
 $R \rightarrow 1111R \mid \varepsilon$
 $P \rightarrow 1P \mid \varepsilon$
 $Q \rightarrow 11Q1 \mid 111 \emptyset$

Question 3

A.

- a) There are three possible parse trees to derive the given string.
 Drawing any two of the following will be correct.



- b) Correct answers include 000111, 001110, 011100, 111000, 110001, 100011. Finding any two of these will be okay. Everything else is incorrect.

B.

- a) & b)

There are three different derivations of abababb.

